SEEDLINE · DESIGN DECISION RECORD

SeedLine TRL 2 review decisions

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DOCUMENT CONTROL

Document ID	SDL-DDR-001	Version	0.2
Status	Draft	Date	2026-10-02
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Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/seedline		
Source	docs/decisions/0001-trl2-review-decisions.md @ 0a57933		

REVISION HISTORY

Version	Date	Author	Change
0.2	2026-10-02	Amish Chadha	Items 10 and 11 decided by Amish on 2026-10-02 (SDL-DEC-001)
0.1	2026-09-25	Amish Chadha	Record Amish's 2026-09-25 decisions on the TRL 2 review and the items that remain open

0001: TRL 2 review decisions

- **Date:** 2026-09-25
- **Status:** accepted (items 1 to 9); item 10 and the plate generator question (item 11) decided by Amish on 2026-10-02 (SDL-DEC-001)

Context

The TRL 2 review note (`docs/REVIEW.md`, session 2026-09-25, /populate) listed ten items as "Proposed, awaiting Amish". On 2026-09-25 Amish reviewed the review points for every portfolio repo and wrote: "proceed with all of your recommendations across all batches. Make sure we don't proceed to TRL 4 on any of them." The same instruction set a portfolio rule that community designs pick co-design partners per area later. The cross-cutting SwapCell decisions do not apply: SeedLine has no battery or electronics.

This record lists what that instruction decides and what it leaves open because there was no recommendation to accept. TRL 4 is on hold by Amish's instruction.

Options considered

The options for each item are in `docs/REVIEW.md` (TRL 2 section) and SDL-PRC-001 v0.2. They are not repeated here.

Decision

Table 1. Decided items.

#	Item	Decision	Where it now lives
1	Metering type	Decided by Amish, 2026-09-25: go with recommendation. Vertical cell plate on a transverse shaft, driven directly by the #35 chain (option A)	SDL-PRC-001 v0.3, `cad/src/model.py`
2	First user group	Decided by Amish, 2026-09-25: go with recommendation. Smallholder field crops first (maize, beans, sorghum, groundnut on 0.75 m rows), with vegetable plates as a second plate set	SDL-PRB-001 v0.3, SDL-REQ-001 v0.3 design case
3	Meaning of the row-spacing kit	Decided by Amish, 2026-09-25: go with recommendation. A telescoping marker arm now; a gang bar for two or three metering units is recorded as a later bed-seeding variant	SDL-PRC-001 v0.3, SDL-REQ-001 R13, `bom/bom.csv` line 14
4	Ratio change	Decided by Amish, 2026-09-25: go with recommendation. Plate cell count plus a 12, 15 or 18 T wheel sprocket, with a spring idler (option A)	SDL-CAL-001 section 2, `bom/bom.csv` line 2
5	R3 long spacings	Decided by Amish, 2026-09-25: go with recommendation. Keep R3 (25 to 400 mm) and add skip-cell plates (plates with one or two cells)	SDL-REQ-001 R3, SDL-CAL-001 section 2
6	Small seed (R5)	Decided by Amish, 2026-09-25: go with recommendation. Specify pelleted seed for 1 to 2 mm seed at first	SDL-REQ-001 R5 (redefined), SDL-CAL-001 section 4
7	Wheel layout	Decided by Amish, 2026-09-25: go with recommendation. Front drive wheel, rear press wheel	SDL-PRC-001 v0.3, `cad/src/model.py`
8	Frame joining	Decided by Amish, 2026-09-25: go with recommendation. Bolted square tube as the default; welding stays an option for workshops that weld	SDL-REQ-001 R18, `bom/bom.csv` line 12, SDL-DWG-001
9	Plate material	Decided by Amish, 2026-09-25: go with recommendation. PETG as the default, ASA where plates sit in strong sun, PLA for trials only	SDL-REQ-001 R6, `bom/bom.csv` line 6

Budget, pitch and problem line: the TRL 2 review proposed no change, so `budget_usd: 400`, the pitch and the problem line stay as they are.

Items that remain open

Table 2. Items open on 2026-09-25, decided on 2026-10-02.

#	Item	Why it is open
10	Co-design partner and region	The TRL 2 review made no recommendation. Under the portfolio rule of 2026-09-25, community designs pick co-design partners per area later, so this stayed open. Decided by Amish, 2026-10-02: a partner already working with smallholder maize and bean farmers on 0.75 m rows on ploughed land; first candidate to approach, CIMMYT's small-scale mechanization work in Eastern and Southern Africa (SDL-DEC-001)
11	Standalone plate generator	SDL-PRC-001 v0.2 listed a parametric plate generator as a suggestion without a recommendation. The TRL 3 model has a parametric plate function (`make_plate` in `cad/src/model.py`); whether to publish it as a separate tool for printers was for Amish. Decided by Amish, 2026-10-02: kept in this repo with a usage note; published as a separate tool only after bench tests confirm its cell sizes (SDL-DEC-001)

Consequences

- R3 is now met on paper through skip-cell plates (SDL-CAL-001 section 2).
- R5 is redefined: seed under 2 mm is sown only as pelleted seed. The paper check shows raw 1 to 2 mm seed would need cells finer than a 0.4 mm FDM nozzle holds.
- The bolted frame adds brackets and bolts, which raises the base parts cost to about $214 and contributes to R17 (about $200) not being met; see SDL-CAL-001 section 11.
- TRL 4 work (bench tests of cell fill and spacing, field trials, build procedures) is on hold by Amish's instruction.